

International AS and A-level

Biology (9610)

BL01 The Diversity of Living Organisms

Report on the examination

June 2024

REPORT ON EXAMINATION: INTERNATIONAL AS BIOLOGY (9610) BL01 JUNE 2024

Overall, students showed a good understanding of most of the topics and nearly all of the questions were attempted. Question 3 was answered particularly well, and question 5 was generally the question that was the most poorly answered.

The main area where students lost marks was in wording their answers well enough to be unambiguous. Key terminology was sometimes lacking eg using the word active site which talking about enzyme function.

QUESTION 01

01.1

Many students (mean of as % of max = 69%) achieved all three marks, although a fair few mixed up the plasmids and DNA. Students that wrote 'genetic material' unqualified for A, or included RNA as an option were not awarded the mark, as this is incorrect.

A surprising number of students included mitochondria and lysosomes as an option.

However, despite these mistakes, few students scored 0 marks.

01.2

The majority of students (mean as % of max = 70%) were able to do the calculation and produce numbers within the correct range. However, a few produced answers with an incorrect order of magnitude or which were not to 2 sig figs, and thus were only awarded 1 mark.

01.3

There were some good answers, with a clear understanding of osmosis, but many students did not realise that this was the basis of the answer, and so could not access mp1 and 2. This meant the overall mean as % of max was 25%.

A surprising number of students seemed confused about the function of a cell wall. Many thought it controlled the water movement in and out of a cell, and some thought that without it the cell would shrivel.

Students needed the idea of the cell expanding and the cell membrane breaking to get mp3, not just 'leaking'.

01.4

Some students did not score this mark due to mentioning the action of penicillin and not mentioning the lack of a cell wall specifically.

01.5

Only about half of the students scored this mark because some answers were not precise enough. Students had to propose equipment that contained smaller intervals. Mention of pipettes or volumetric flasks were not awarded as they did not fall into this category, and measuring cylinders were only awarded if they were specified to be 100cm³ or below for the same reason.

01.6

Many students forgot to mention enzymes specifically or only talked of their denaturing at high temperatures, and not the idea of optimising enzymic activity.

Some students did not link the optimal activity of the enzymes to increased growth rate of the prokaryotes and so failed to score the second mark.

01.7

This question has an evaluate command word and so students needed to phrase mp1 as an argument in support of the conclusion in order to obtain the mark. They also had to link it to the idea of effectiveness.

The most common marking points awarded were 2 and 3, and most students were able to complete the calculations correctly. However, some students did not achieve mp3 as they calculated the difference between the sizes and not how many times bigger, they are from each other, or claimed that 3.14 is twice the size of 0.79.

Few students were awarded mp4, as most mentioned using a different type of prokaryote, but the experiment was only testing one type.

QUESTION 02

02.1

Many students scored all 4 marks (overall mean as % of max = 86%), but a surprising number were not able to identify the hydrogen bond, or called the organic base a nucleotide.

Some students correctly identified J as an organic base, but then went on to elaborate that it was guanine or cytosine, so could not be awarded the mark due to this being incorrect.

02.2

Most students scored 2 marks here. Some did not put their answer into standard form and so lost a mark. Some achieved the correct answer, but then represented it incorrectly eg 1.2×10^8 , and so could not be awarded any marks.

02.3

Most students scored both marks as they remembered that mRNA has uracil in place of thymine.

02.4

The majority of students scored this mark, although some gave incorrect information (eg the DNA from prokaryotes is always shorter) or details not relevant to the question (eg about the process of transcription or translation, rather than the structure of the mRNA itself). A surprising number of students thought only prokaryotic mRNA contains uracil instead of thymine.

2.5

Most students scored all the marks available (overall mean as % of max = 81%), with only a few mixing up their tests and results.

QUESTION 03

03.1

Most students were able to achieve 2 marks for this question, and many achieved all 4 (overall as % of max = 84%). The most common responses were to correct the use of ink for drawing the line and to suggest using pencil instead. Second most common was to notice that the sample was below the solvent line, but some students did not achieve the marks for this due to confused wording.

03.2

Many students correctly identified cysteine and were able to support this with a R_f value of 0.4. Some students performed the calculation incorrectly and identified the sample as alanine.

03.3

Many students scored both marks for this question, with mp2 being the most commonly awarded marking point. Some students gave answers which were too ambiguous to be awarded eg 'use a stronger' solvent did not score mp2, as it could also have referred to the same solvent, just at a higher concentration.

A fair number of students suggested changes that would not affect the result, eg more concentrated sample or repeating the experiment with no changes.

Some students suggested answers from other parts of the specification, eg DNA hybridisation and immunological comparisons, which were not awarded as they were not relevant.

QUESTION 04

04.1

Around half the students were able to score both marks on this question. Common mistakes were just repeating the word fluid for mp1, rather than giving an explanation and also not mentioning both proteins and phospholipids for mp2.

04.2

For the first gap, answers must be phospholipid as the molecule not just tails. A significant number of students mentioned membrane proteins for this gap, but this was not awarded as non-polar substance transport was involved.

There seemed in general to be lots of confusion over the role of cholesterol.

04.3

Very few students scored both marks (overall mean as % of max = 33%). Many students lost marks because they did not make it clear that their answers were not referring to the cell surface membrane. A surprising number of students thought that the plasma membrane stopped water from passing through it entirely, or thought that it provides strength.

Some students lost a mark because they wrote the same idea twice, just referencing 2 different organelles and so could only be awarded once.

04.4

Relatively few students scored both marks (overall mean as % of max = 22%). Most identified the type of transport as diffusion and talked about equilibrium being reached on both sides of the

membrane. This could not be the case because as the amount of substance P outside the cell increase, the rate of simple diffusion could not be expected to level off.

04.5

Most students scored both these marks (overall mean as % of max = 84%).

QUESTION 05

05.1

A fair few answers were too vague to score full marks. 'Working muscles' was not considered a substitute for respiring muscles, and the idea of oxygen being supplied needed to be present.

Many students did not clearly link the idea of more aerobic respiration to the long-distance athlete and therefore did not obtain the second mark. There needed to be a clear link to how the increased aerobic respiration potential would sustain long distance running, ie less lactic acid build up or the ability to exercise for longer. 'Supplying more energy' was not considered the same as 'more energy produced in muscles'.

05.2

A number of students did not appreciate the fact that the red blood cell measurement was its diameter, and the capillary measurement was the radius, and so did not score mp1. The most common mark awarded was mp 3. Few were awarded mp 4.

There was some vague language used here eg easier diffusion is not the same as faster diffusion, and many students thought that the capillaries were able to contract and dilate.

05.3

Relatively few students scored both marks. At least half the students did not score the first mark as the words 'haem group' was not enough to secure the mark, as this was already mentioned in the stem. Marks were only awarded for iron ions, not just mention of iron.

Some students were not awarded mp2 because they did not refer to oxygen being unloaded as well as loaded.

Some students misinterpreted the question and instead wrote about the conformational shape changes caused when oxygen binds to haemoglobin.

05.4

The vast majority of students scored both marks for this calculation, with the ones that only scored 1 or 0 marks generally only marginally out eg giving answers of 58% or 3.8.

05.5

A number of students showed poor understanding and thought that a shift to the right meant that more oxygen could be loaded at the lungs and therefore be available to respiring tissues. To score both marks, answers needed both the idea of faster oxygen release (not just lower oxygen saturation on the haemoglobin) and also the idea of more oxygen available to respiring tissues.

QUESTION 06

06.1

Most students scored both marks (overall mean as % of max = 71%). Those that lost marks tended to do so in the second column: some thought that fish was a kingdom, or that fish are mammals.

06.2

A significant number of students lost marks due to lack of appropriate terminology. The idea of being 'closely related' or 'sharing common ancestors' was needed to get the marks. Students that instead assigned a taxonomic group to each species also lost the marks, as there was no information about how to do this correctly in the question.

Some students misunderstood the phrase recent common ancestor and thought that none of the pike shared a common ancestor.

06.3

Many students lost a mark for this question because they did not word their answers comprehensively enough. It was not enough to talk of breeding the species together to see if they produce fertile offspring because the question was asking how to check if they were new species. The results of the breeding experiment had to be clearly linked to the conclusion drawn from them to achieve the mark.

Ideas of DNA/mRNA/protein sequencing were only awarded if the idea of a comparison was used.

06.4

Many students lost marks on this question due to lack of clarity in their answers. Mp 1 was often lost due to the answer potentially being interpreted as a definition of species richness eg 'total numbers of all the species'.

Mp2 was scored much more often than mp1.

06.5

Most students scored the second mark for the idea that random sampling removes bias, but about half did not score the first mark for the idea of a large sample size. Answers that included reference to a mean value were ignored, as a mean is not necessary when measuring the diversity index. Answers that referenced the idea of reducing the effect of chance were also ignored, as they were generally not specific enough. There needed to be the idea of reducing the effect of anomalies, not just reducing anomalies, as large data sets do not eliminate anomalies.

QUESTION 07

07.1

The vast majority of students achieved this mark (overall mean as % of max = 85%) and knew the name of the model.

07.2

Again, most students scored this mark (overall mean as % of max = 90%), despite there being some unusual spellings of hydrolysis and some instances of 'condensation reactions'

07.3

Mp1 needed the idea of 2 active sites being complementary to the same substrate to be awarded the mark. Many students put forward answers about the enzymes having the same shape active sites, or being able to change shape around the substrate, but these were only awarded if it was clear there were 2 active sites and one substrate. Overall mean as % of max = 18%).

07.4

Many students scored both marks here, but some lost marks through using imprecise language. The active site had to be mentioned to score either mark, alongside the idea of the substrate now being complementary. The word 'fit' was accepted here as a substitute for this, but the word 'bind' was not, as it didn't convey the concept adequately.

The students needed to convey the idea that the substrate is binding to both the cofactor and the original active site of the enzyme to get mp 2.

07.5

Most students were able to score full marks for this question (overall mean as % of max = 77%). Those that missed the marks did so because they were not clear that molecule Z was not bonding onto the active site, and also because they mis-named the non-competitive inhibitor.

07.6

Comparatively few students knew to increase the enzyme concentration. There were many suggestions about changing molecule Z and these were mostly awarded if they seemed feasible. The suggestions had to be something that could practically be done to the pre-existing experiment. Answers that involved something binding to the allosteric site were not awarded as these could still alter the shape of the active site.